

Research and Applications

Improving clinical decision support through interpretable machine learning and error handling in electronic health records

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Abstract

Objective: To develop an electronic medical record (EMR) data processing tool that confers clinical context to machine learning (ML) algorithms for error handling, bias mitigation, and interpretability.

Materials and Methods: We present Trust-MAPS, an algorithm that translates clinical domain knowledge into high-dimensional, mixed-integer programming models that capture physiological and biological constraints on clinical measurements. EMR data are projected onto this constrained space, effectively bringing outliers to fall within a physiologically feasible range. We then compute the distance of each data point from the constrained space modeling healthy physiology to quantify deviation from the norm. These distances, termed “trust-scores,” are integrated into the feature space for downstream ML applications. We demonstrate the utility of Trust-MAPS by training a binary classifier for early sepsis prediction on data from the 2019 PhysioNet Computing in Cardiology Challenge, using the XGBoost algorithm and applying SMOTE for over-coming class-imbalance.

Results: The Trust-MAPS framework shows desirable behavior in handling potential errors and boosting predictive performance. We achieve an area under the receiver operating characteristic curve of 0.91 (95% CI, 0.89–0.92) for predicting sepsis 6 hours before onset—a marked 15% improvement over a baseline model trained without Trust-MAPS.

Discussions: Downstream classification performance improves after Trust-MAPS preprocessing, highlighting the bias reducing capabilities of the error-handling projections. Trust-scores emerge as clinically meaningful features that not only boost predictive performance for clinical decision support tasks but also lend interpretability to ML models.

Conclusion: This work is the first to translate clinical domain knowledge into mathematical constraints, model cross-vital dependencies, and identify aberrations in high-dimensional medical data. Our method allows for error handling in EMR and confers interpretability and superior predictive power to models trained for clinical decision support.

Key words: interpretable machine learning; clinical decision support; clinical data preprocessing; outlier detection; context-aware algorithms.

Background and significance

Complex decision-making in time-critical domains like the intensive care unit (ICU) involves careful attention to information from several data sources like the patient’s medical history, and their current clinical tests. Trained clinicians comprehend the limitations of each data type in a given clinical situation and evaluate the entirety of this data, easily recognizing conflicting information. Thus, they continually navigate a multidimensional feature space grounded in physical and biological constraints. Using their domain knowledge, they can pick out outlier data points and can attribute them to errors in the data.

There is growing interest in leveraging machine learning (ML) to support clinical decision-making in tasks like ICU mortality prediction and early sepsis detection.¹ ML models are trained on retrospective patient data stored in electronic medical records (EMRs). However, EMRs often contain missing or erroneous data.^{2,3} While clinicians can interpret and amend discrepancies based on medical expertise, ML pipelines do not have the clinical context to be able to make similar corrections. In this work, we address the challenge of embedding domain knowledge into ML pipelines to create context-aware algorithms that can automatically detect data errors, thereby reducing model bias. We also introduce a

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feature engineering method using domain knowledge that enhances both interpretability and predictive performance. We present this approach as a plug-and-play preprocessing tool for high-fidelity ML, called *Trust-MAPS*—Trustable Real-time and Uncertainty-managed *ST*reaming ML Algorithms using Projections for Safety critical diagnosis.

Error identification in large medical datasets has traditionally relied on outlier detection methods, such as inter-quartile range tests,⁴ mean-variance tests,⁵ and k -Nearest Neighbour algorithms.⁵ Errors are typically handled using supervised learning techniques like regression models.^{4,6,7} However, these methods do not incorporate clinical context into the data processing pipeline. Trust-MAPS is a constrained optimization approach for detecting potential data errors and integrating clinical knowledge by translating complex biological and physiological constraints on vitals and lab values into high-dimensional mathematical constraints. The “projection” of data derived from the EMR onto a sub-space defined by feasible domain constraints allows us to bring outliers back into the physically possible range of values, consequently bolstering the reliability of subsequent data-driven learning tasks. We also design a novel set of features, called *trust-scores*, by computing the distance between a patient’s clinical state—defined by vital signs and laboratory values—and the constrained space representing normal ranges for healthy patients. Trust-scores can offer interpretable insights to explain model predictions. We validate our framework by applying it to ML models trained on the PhysioNet Computing in Cardiology Challenge 2019⁸ for the early prediction of sepsis. While the primary focus in this study is sepsis prediction, the projection-based framework of Trust-MAPS is versatile and can generalize to other clinical decision support models.

Recent advances in sepsis prediction leverage various ML techniques, including gradient-boosted trees,^{9,10} deep learning architectures such as LSTM,^{11–13} and hybrid approaches that combine these methods.¹⁴ However, challenges persist due to the susceptibility of these algorithms to biases and systemic errors present in training data, which often lead to high false alarm rates and clinician fatigue.^{13,15} To mitigate these issues, researchers have aimed to improve ML model robustness by incorporating clinical knowledge,^{16,17} addressing uncertainties,¹⁸ and improving interpretability.^{11,19,20} Some approaches for adding clinical context include knowledge graphs,¹⁶ risk factor-based regularization,²¹ and expert-defined rule integration.¹⁷ Our method builds on the latter by introducing a flexible framework that can handle high-dimensional data, accommodate range-based constraints, and capture cross-feature interactions. Methods to confer interpretability to ML algorithms include attention mechanisms for deep learning,^{11,12} SHAP values for traditional classifiers,⁹ rule-based models with target-defined separation thresholds,^{22–24} and subgroup discovery.²⁵ Our results show that the Trust-MAPS framework complements these approaches well and can be used as preprocessing step to any downstream ML pipeline.

Thus, we present Trust-MAPS as a data preprocessing module with the following functionalities: (i) handling potential errors in EMR data, (ii) integrating clinical context into ML pipelines through expert-defined constraints on clinical variables, and (iii) engineering a new set of interpretable features, termed *trust-scores*, that quantify a patient’s feature-specific deviation from healthy physiology.

Methods

A *projection* is a fundamental operation that arises in many different fields, ranging from architecture to ML.²⁶ The projection of a given point $x \in \mathbb{R}^n$ in high-dimensional space to some set $S \subseteq \mathbb{R}^n$ is the “closest” point in the set S to x . For example, a point in 3-dimensional space (eg, a ball in the air) will project onto its shadow on the floor.

The key idea of Trust-MAPS is to model prior knowledge of feasible data values as high-dimensional mathematical constraints. These constraints could be upper and lower bounds on feature values, physically possible rate of change in recorded values, or conditional interactions between variables. We define the following 2 sets of clinically interpretable constraints:

- *Physical constraints* (set P): These constraints define biologically possible bounds on values and rates of change of clinical data of critically ill ICU patients, as well as dependencies between various clinical measurements. For example, the physical range for heart rate is 30–200 beats per minute. These constraints also capture the rate of change of various vitals and lab values, as well as dependencies across different values, eg, if *bicarbonate* is less than 10, then *base excess* must be at most 0 (detailed in Section S6.2.1).
- *Normal constraints* or *homeostasis constraints* (set N): These are constraints on the expected range of values of a healthy patient’s data, for example, the normal range for heart rate is 60–90 beats per minute, along with rate constraints and dependencies across lab values and vitals (detailed in Section S6.2.2).

We are given observed, temporal patient data, $\text{data}[v, t]$ from the EMR, where $v \in V$, the set of all vitals and lab values in consideration. The projection, $x[v, t]$ of $\text{data}[v, t]$ onto the set of constraints P is a way to determine which values in $\text{data}[v, t]$ are erroneous and to bring these values into a clinically feasibility range. The goal is to prevent the ML model from learning incorrect patterns from erroneous values and to avoid scenarios in which single, abnormal events significantly impact decision-making. On the other hand, the distance of the projected data $x[v, t]$ to constraints capturing normal physiology (set N) are informative of how sick the patient is and enrich the feature space for downstream ML. An example of the application of these “projection” steps is shown in Figure 1.

The length of temporal patient data varies with the duration of ICU stay. To apply Trust-MAPS, it is necessary to create a data point of fixed dimensions. One way would be to aggregate feature recordings every hour and treat each hour as a separate data point. However, temporal trends, which can be informative in clinical decision-making, are lost. For this study, we divide patient data into fixed time intervals, referred to as *sub-patients*, with each *sub-patient* treated as an individual data point. A *sub-patient* provides a snapshot-in-time of the patient’s overall ICU stay. For a fixed time interval n and number of clinical variables f , we generate snapshots of size $n \times f$, unroll each clinical feature, and create 1-dimensional sub-patients of length $n * f$. A detailed rationale and methodology for creating *sub-patients* is provided in Section S6.1.3. For the results in this paper, we use 6-hour intervals with a 3-hour overlap and include similar plots for other intervals and overlaps in Section S10.3. Data are pre-processed by imputing missing values (Section S6.1) and then applying min-max normalization (Section S6.1.2).

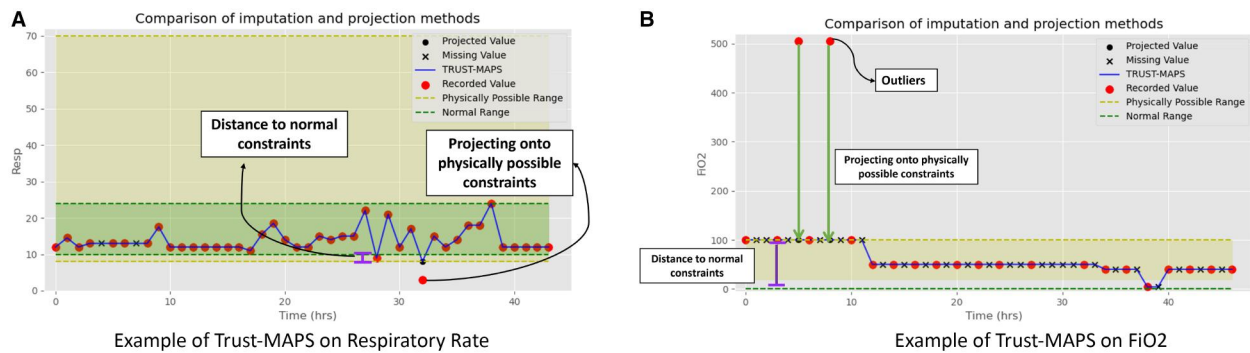


Figure 1. A visualization of the Trust-MAPS processing on clinical data. In the first step, outlier data points that lie outside the physically possible constraints are projected to lie within the constraints. This is to prevent the machine learning model from learning incorrect patterns from erroneous data and mitigate potential bias in the downstream learning task. In the second step, the distance to the normal constraints is calculated to provide a cohesive number that quantifies how deviated the patient is, in any particular interval of time (*sub-patient*), from normal physiology.

The Trust-MAPS pipeline is a 2-step process. First, we project patient data onto set P , as described in detail in Section S6.2.1. The resulting data are considered free of outliers and erroneous values. Next, we calculate the Euclidean distance of the corrected data to its projection on the decision set N . For a given variable v in a *sub-patient* of time-interval n , if $v^P = [v^P_1, v^P_2, \dots, v^P_n]$ represents the vector of values of v recorded during that *sub-patient* interval and projected onto P , and $v^N = [v^N_1, v^N_2, \dots, v^N_n]$ represents the projection of v^P onto N , then the Euclidean distance $\|v^P - v^N\|$ is what we call the distance-to-normal or “trust-score” for v in that *sub-patient*. Trust-scores give us a trustworthy and interpretable measure of how sick a patient might be. They reduce a sequence of n values recorded over a period of time to a single number that quantifies the deviation from healthy physiology, with respect to each clinical variable [eqn (2)]. We subsequently use trust-scores as input features for training a ML model for prediction of sepsis. This process is described in Section S6.2.2, see Figure 2. The complete list of upper and lower bounds on clinical data used to define N and P , and the rate constraints, can be found in Section S9, and the detailed formulation of our *if-then-else* constraints can be found in Section S7. These clinical constraints were formulated by a panel of 3 board-certified physicians, following a consensus achieved through collaborative discussions.

Results

Data description and summary of clinical characteristics

We assess the performance of Trust-MAPS on a publicly available dataset released as part of the PhysioNet Computing in Cardiology Challenge 2019.⁸ (The data can be accessed at <https://physionet.org/content/challenge-2019/1.0.0/>.) The data consist of 6 demographic features, 8 vital signs, and 26 laboratory variables aggregated into hourly bins and recorded over a patient’s stay in the ICU. Thirty clinical variables from the PhysioNet dataset were used for building our model. These are enlisted along with their expert-defined physical and normal range constraints in Table S4 of Section S9. We compute the SIRS (Systemic Inflammatory Response System)²⁷ and SOFA (Sepsis Related Organ Failure Assessment)²⁸ score for benchmarking our ML algorithm. (We also use SOFA and SIRS as predictors in our ML model to improve prediction power and interpretability.) Each patient

record has an associated time-dependent “Sepsis Label” column which is set to 1 six hours before the onset of sepsis. The time at which the patient develops sepsis is decided according to the Sepsis-3 definition,²⁹ which is given in Section S8. The data consisted of 40 336 patients, out of which 2932 patients (7.2%) developed sepsis. After standardizing to fixed “sub-patient” time windows, we obtained a total of 463 693 sub-patients, out of which 9646 (2%) correspond to sepsis patients. Table S5 in Section S9 details the complete descriptive statistics of the dataset.

Data correction by projection onto physical constraints

Projecting data onto P , defined as a “physical projection,” can bring outlier data points to a value within the physically possible constraints. The constraints we define account for their minimum and maximum values, maximum rate of change values, and interactions between variables. Figure 3 illustrates the physical projection step demonstrated through constraint on the relationship between mean arterial pressure (MAP), systolic blood pressure (SBP), and diastolic blood pressure (DBP) (Section 6.2.1, constraint #4). In this case, there was no recorded value of DBP in the EMR. Trust-MAPS first imputes DBP to a constant value equaling the mean of the “normal” range of values, ie, 70. However, the physical projections step captures the interaction between DBP, SBP, and MAP and autocorrects DBP accordingly. We refer the reader to Figures S9-S12 in Section S10 for more interesting examples of data correction related to phosphate, respiration, base excess, bicarbonate, and lactate.

We compare Trust-MAPS with alternate methods of imputation, as detailed in Section S10.6. Particularly, we test Trust-MAPS against MICE,⁷ a commonly used imputation technique for healthcare data, along with alternate formulation of the optimization objective used for projections. The behavior of different imputation strategies is displayed in Figure 3C, with more examples in the Figure S21. We can see that the linear imputation and projections approach used by Trust-MAPS shows desirable behaviors in correcting outlier values and faithfully imputing missing values.

A caveat to note is that the distance of a data point from the constraint set P does not always reflect the true extent of error in the variable. For example, a misplaced decimal point can result in a large distance. However, any non-zero distance still signals potential data errors and informs confidence in

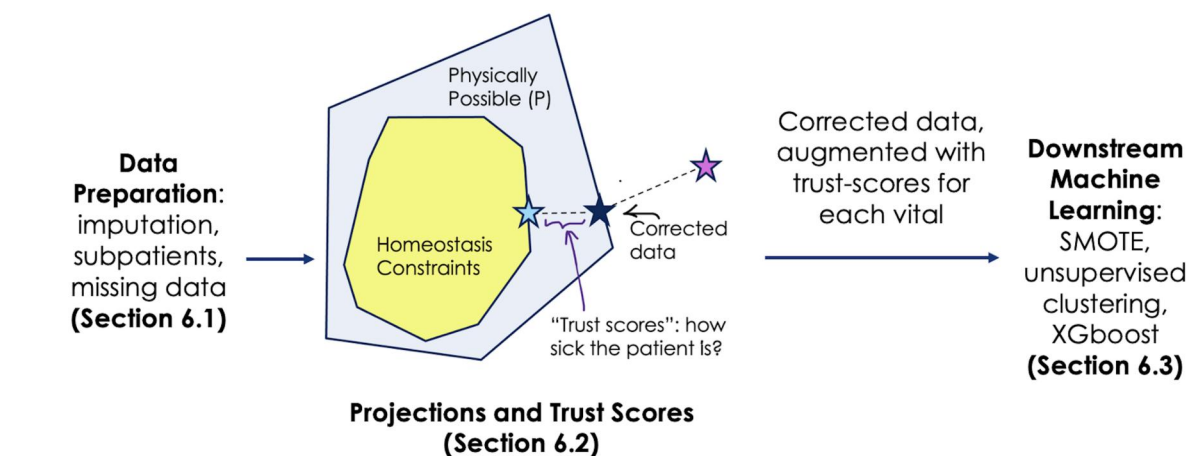


Figure 2. Trust-MAPS augment the “corrected data” computed using projections onto physical constraints P , with “trust-scores” which capture the distance of the corrected data from homeostasis constraints. This augmented data are then used in downstream ML for predictions, which significantly improves the performance. There are a number of steps needed to normalize the length of the stay of patients, impute missing values in the data to get the new patient data in standardized form, and then data augmentation with trust-scores for each patient vital (see Figure S7 in Section S6), before using projections. The pipeline is very general and can be used for more advanced predictive methods (eg, those that process clinical notes).

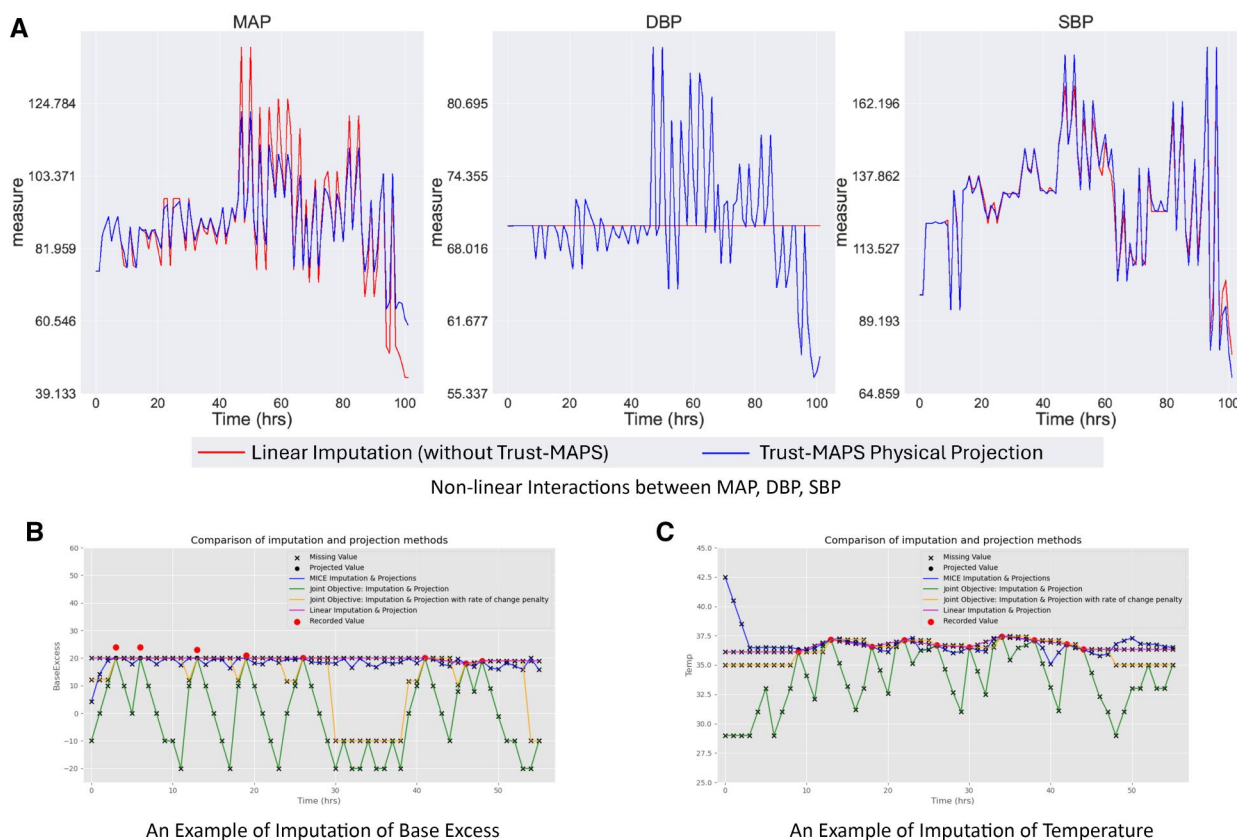


Figure 3. An illustrative example of data correction by projection on the Physical Constraints Set P demonstrated through (A) the constraint on the relationship between MAP, DBP, and SBP, where the imputed values of DBP default to the average value but the projected values satisfy the interactions between these vitals given by constraint #4 in Section S6.2.1. (B) A comparison of imputation of Base Excess by methods described in Section S10.6. The Trust-MAPS approach is labeled “Linear Imputation and Projections.” (C) A comparison of imputation of temperature.

subsequent model predictions. (Most laboratory values are transmitted automatically from lab machines to the EMR. Thus, detected errors might indicate issues in sample collection or laboratory procedures, which we do not address in this work).

Trust-MAPS case study: sepsis prediction Superior classifier performance for sepsis prediction

The ML pipeline used in this study was designed to combat the severe class imbalance observed in the data and to maximize precision (Section S6.3). Given the highly heterogeneous

nature of patient data, we follow the Cluster-then-Predict approach³⁰ to maximize performance. We perform unsupervised *k*-means clustering with 25 clusters on the training data and train a different XGBoost³¹ on each cluster. Details regarding the choice of the number of clusters and methods of analysis are described in Section S6.3, and the results are presented in Figure S13 in Section S10.

The baseline for comparison of classification performance for sepsis prediction is a model trained on the dataset after linear imputation and without Trust-MAPS. We then train 2 separate models on data obtained after applying the first step of Trust-MAPS (physical projection) and the second step of Trust-MAPS (normal projection), respectively. As an additional comparison, we train a ML model after applying MICE.

We refer the reader to Table 1 and Figure 4 for the results comparing the performance of the data correction and imputation techniques. The model trained on data with Trust-MAPS achieves an AUC of 0.901, approximately a 15% improvement over the baseline. The model trained with Trust-MAPS is also better calibrated, as is evident from the reliability curve and brier score results detailed in Section S11.

In Figure 5, we plot a histogram depicting the distribution of the number of hours before sepsis onset that our algorithm raises its first alert for a patient. In Figure 5A, we observe that most correct sepsis predictions were made 8–9 hours before the patient developed sepsis, which is a *significant* lead time for clinical action. In Figure 5B, we observe that the model trained without Trust-MAPS has a more widely spread distribution of sepsis detection times, which is not a desirable property for a clinical decision support model. A model that raises alerts for sepsis well before the prediction window of 6–12 hours can be prone to false positives and can lead to alarm fatigue in the clinicians, whereas sepsis alerts raises too close to the time of onset may not provide enough time to take timely action.

The PhysioNet Challenge⁸ used a *utility score* metric to evaluate sepsis prediction models, which accounts for the

timing between predicted and actual sepsis onset. This score penalizes predictions that are significantly early or late. To validate Trust-MAPS as a flexible preprocessing module compatible with any ML pipeline, we compared its performance against the sepsis prediction model developed by Yang et al.,⁹ which used XGBoost with Bayesian Optimization and achieved the highest average utility scores of 0.522 on the PhysioNet dataset. First, we replicate their pipeline to establish a baseline, then apply it to data processed with Trust-MAPS. We achieved an average utility score of 0.566—an 8% improvement over the baseline of Yang et al.⁹ The results are documented in Table 2. We can clearly see a boost in performance over the baseline.

The [Supplementary Material](#) contains details of our experiments with other classification algorithms (Sections S10.4 and S10.5), different sub-patient durations (Section S10.3), and alternate imputation strategies (Section S10.6).

Feature importance and model interpretability

A trained XGBoost model automatically calculates features of the maximum importance for the predictive modeling problem. Feature importance gives us insights into the clinical variables that have the maximum discriminatory power for sepsis prediction. It is calculated as the number of times a variable is selected for splitting a node, weighted by the squared improvement to the model as a result of each split, and averaged over all trees.³² We calculate the average feature importance for each variable averaged over the 6 hours sub-patient time window for our analysis. In Figure 6, we extend our analysis of feature importance to models trained on each of the 25 clusters identified by the *k*-means clustering step of our algorithm design, and plot the top 10 most important features for the prediction problem for each cluster. We find that the trust-scores were some of the most critical predictors in our ML model and imperative for effective decision-making. Trust-scores confer a natural interpretability to model predictions. This is highlighted in Table 3, where a trained clinician adjudicated the basis for sepsis classification for each cluster based on the variable specific trust-scores

Table 1. Confidence intervals for the sensitivity, specificity, precision, AU-ROC (area under the receiver operating characteristic curve), AU-PRC (area under the precision-recall curve), and *f*-score of the sepsis prediction algorithm over 50 iterations.

Dataset	Method	Sensitivity	Specificity	Precision	AU-ROC	AU-PRC	<i>f</i> -score
Train set	Trust-MAPS normal projection	0.804 ± 0.001	0.999 ± 0.00002	0.974 ± 0.001	0.961 ± 0.001	0.857 ± 0.001	0.881 ± 0.003
	Trust-MAPS physical projection	0.793 ± 0.005	0.999 ± 0.0003	0.956 ± 0.002	0.950 ± 0.002	0.846 ± 0.004	0.867 ± 0.003
	MICE imputation	0.579 ± 0.001	0.996 ± 0.0001	0.245 ± 0.002	0.891 ± 0.006	0.349 ± 0.004	0.344 ± 0.003
	Baseline (without Trust-MAPS)	0.614 ± 0.003	0.951 ± 0.003	0.538 ± 0.012	0.828 ± 0.012	0.199 ± 0.031	0.552 ± 0.030
	SOFA scores	0.168 ± 0.012	0.896 ± 0.001	0.054 ± 0.021	—	—	0.081 ± 0.081
Test set	Trust-MAPS normal projection	0.633 ± 0.018	0.998 ± 0.0003	0.951 ± 0.004	0.907 ± 0.008	0.707 ± 0.017	0.760 ± 0.012
	Trust-MAPS physical projection	0.622 ± 0.011	0.993 ± 0.0009	0.910 ± 0.005	0.882 ± 0.010	0.684 ± 0.017	0.739 ± 0.015
	MICE imputation	0.370 ± 0.002	0.958 ± 0.001	0.165 ± 0.001	0.833 ± 0.002	0.248 ± 0.001	0.228 ± 0.012
	Baseline (without trust-MAPS)	0.445 ± 0.015	0.949 ± 0.003	0.499 ± 0.007	0.785 ± 0.019	0.176 ± 0.012	0.435 ± 0.014
	SOFA scores	0.152 ± 0.020	0.897 ± 0.001	0.05 ± 0.025	—	—	0.075 ± 0.074

We benchmark the performance of the machine learning model for sepsis prediction at each step of the Trust-MAPS pipeline against the performance of models trained without Trust-MAPS. Additionally, we compare the performance on the same machine learning pipeline on data imputed using MICE. We also benchmark against using the SOFA score as an indicator of sepsis. For this purpose, a SOFA score greater than or equal to 2 indicates that the patient has developed sepsis.²⁸

The bold values represent the best performing methods for each performance metric.

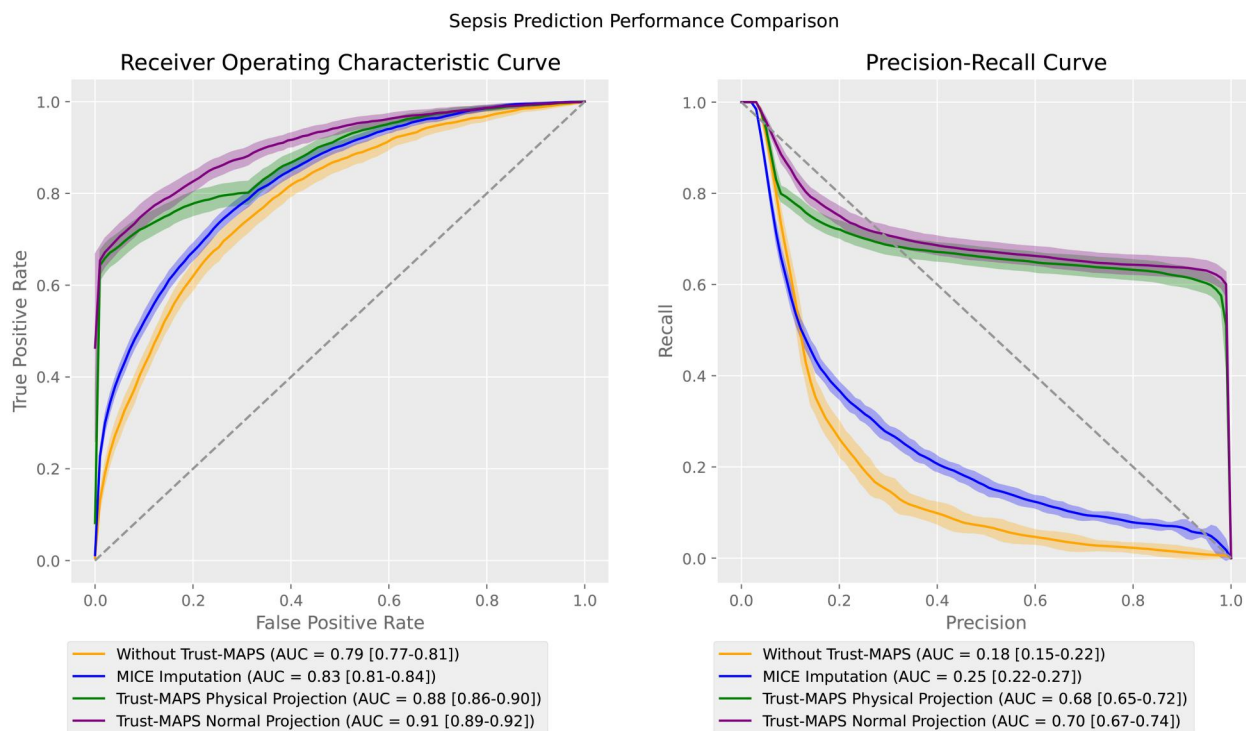


Figure 4. Left: A plot comparing receiver-operating characteristic curves for sepsis prediction machine learning model on the dataset processing with each step of the Trust-MAPS process, and baseline models trained without Trust-MAPS. Right: A plot comparing precision-recall curves for sepsis prediction machine learning model on the dataset processing with each step of the Trust-MAPS process, and baseline models trained without Trust-MAPS.

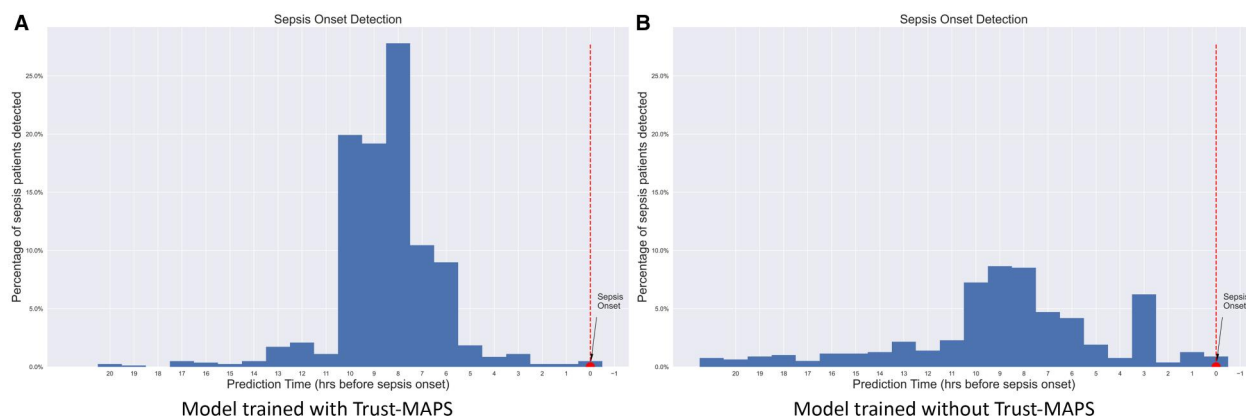


Figure 5. Plot obtained by calculating how many hours in advance our ML model correctly predicts the onset of sepsis for every sepsis patient in the test set. Here, the time of sepsis prediction is the end of the *sub-patient* time window at which our model raises the first sepsis alert for the patient. It is desirable to predict sepsis 6-12 hours before onset. Raising an alert too soon may be an outcome of a highly sensitive model prone to false positives, which can lead to alarm fatigue.

that were most important for classification in each cluster (Figure 6).

Discussion

We introduce Trust-MAPS, a method for preprocessing EMR data that integrates clinical domain knowledge into data-driven pipelines using principles of constrained optimization. We demonstrate the framework's utility by training a supervised ML classifier on temporal patient data, encompassing clinical laboratory values and vital signs. In healthcare applications, where predictive algorithms alert providers to

potential medical conditions, the positive predictive value is crucial. Too many false positives could lead to system disengagement by healthcare staff, a challenge with current ML algorithms in practice.¹⁵ Therefore, we consider precision an essential metric for classifier performance. The model trained with Trust-MAPS achieved an AUC of 0.91 (95% CI, 0.89-0.92) and a precision of 0.95 (95% CI, 0.94-0.95) for predicting sepsis 6 hours before onset. This represents a substantial improvement over the baseline model trained on data without Trust-MAPS, which achieved an AUC of 0.79 (95% CI, 0.77-0.81) and a precision of 0.50 (95% CI, 0.49-0.51). Additionally, a model trained after applying only the first

Table 2. Performance results with confidence intervals for the accuracy, AU-ROC (area under the receiver operating characteristic curve), AU-PRC (area under the precision-recall curve), utility score,⁸ and *f*-score on sepsis prediction combining Trust-MAPS as a preprocessing tool with the prediction algorithm proposed by Yang et al.⁹

Method	Utility	Accuracy	<i>f</i> -score	AU-ROC	AU-PRC
Trust-MAPS preprocessing (Normal Proj.) with Yang et al. ⁹	0.566 ± 0.005	0.889 ± 0.004	0.194 ± 0.005	0.905 ± 0.013	0.357 ± 0.007
Trust-MAPS preprocessing (Physical Proj.) with Yang et al. ⁹	0.565 ± 0.004	0.882 ± 0.065	0.187 ± 0.004	0.903 ± 0.003	0.355 ± 0.011
Baseline ⁹ (Without Trust-MAPS)	0.522 ± 0.004	0.845 ± 0.078	0.151 ± 0.005	0.894 ± 0.003	0.193 ± 0.018

The bold values indicate the method (rows) with the best performance for each metric (columns).

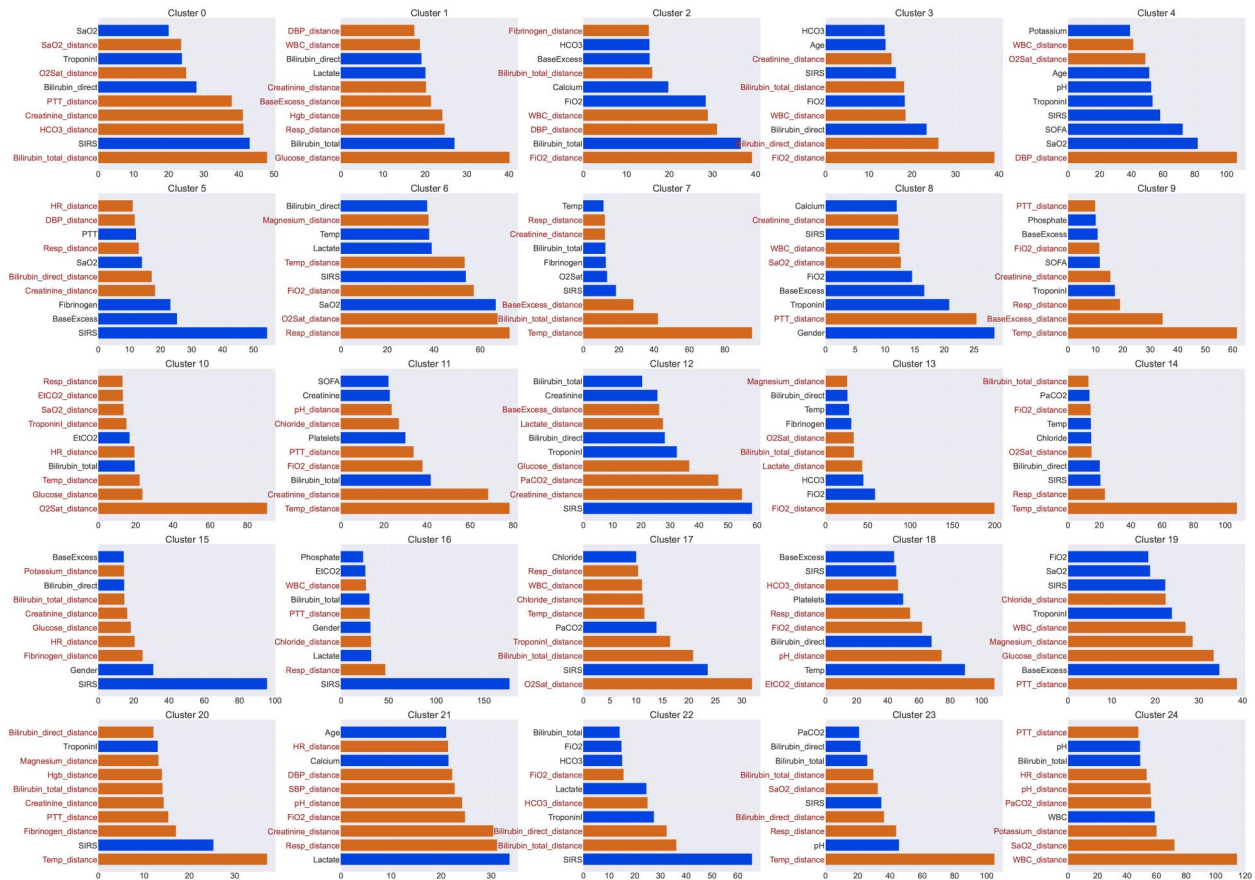


Figure 6. This figure shows the cluster-wise features of importance for XGBoost for each cluster identified by *k*-means clustering. The trust-scores (labeled “distance” variables), generated by calculating the distance of a sub-patient from the set of normal (or healthy) constraints, are marked in orange. We can see that these trust-scores play an important role in teaching the model to predict sepsis.

step of the Trust-MAPS process (the physical projection) achieved an AUC of 0.88 (95% CI, 0.86-0.90) and a precision of 0.91 (95% CI, 0.90-0.92) in the same prediction setting.

Our results underscore 2 key points. First, the “physical projection” step alone improves classification performance, reinforcing the bias reducing capability of the error-handling step. Second, the trust-scores calculated by the “normal projection” serve as clinically interpretable features that enrich the feature space with predictive power, resulting in superior classification performance.

Clinical decision support models that accurately assess sepsis risk can be instrumental in reducing poor outcomes.³³ As a result, many sepsis prediction algorithms have been developed across diverse data types and sources. For example,

Goh et al.³⁴ developed an ML algorithm achieving an AUC of 0.90 and a precision of 0.82 for sepsis prediction 6 hours before onset. However, their results were based on a different dataset and incorporated clinical notes. To better assess Trust-MAPS, we benchmark our results against approaches that report outcomes on the PhysioNet Challenge dataset. The winning team¹⁰ of the challenge reports an average training set utility score of 0.430 and a test set utility score of 0.360. However, this comparison is limited, as the test utility scores were calculated on a hidden dataset that we did not have access to. Therefore, we validate Trust-MAPS by testing it against the method proposed in Yang et al.,⁹ which uses XGBoost with Bayesian Optimization to train a sepsis prediction model, achieving an average utility of 0.522. Higher predictive performance has been reported in works,^{11–13} but

Table 3. The clinical interpretation relating to organ dysfunction profiles of the clusters using their most predictive features (ie, specifically the trust-scores for various lab values and vitals).

Cluster	Clinical interpretation
0	Classification driven by subtle multiorgan derangement, with liver, SIRS, and kidney predominance
1	Classification driven by subtle multiorgan derangement with predominant metabolic disturbance (glucose derangement)
2	Classification driven by subtle multiorgan derangement, with respiratory, hemodynamic (vascular tone), and inflammatory predominance
3	Classification driven by subtle multiorgan derangements, with respiratory and liver predominance
4	Classification driven by subtle multiorgan derangements, with hemodynamic predominance from changes in vascular tone
5	Classification driven predominantly by inflammatory derangements (SIRS)
6	Classification driven by subtle multiorgan derangements, with respiratory and inflammatory predominance
7	Classification driven by subtle multiorgan derangements, with inflammatory and liver predominance
8	Classification driven by subtle multiorgan derangements, with coagulopathy predominance
9	Classification driven by subtle multiorgan derangements, with inflammatory and acidosis predominance
10	Classification driven by subtle multiorgan derangements, with respiratory predominance due to oxygen measurement
11	Classification driven by subtle multiorgan derangements, with inflammatory and kidney predominance
12	Classification driven by subtle multiorgan derangements, with SIRS, kidney, and metabolic disturbance predominance
13	Classification driven by subtle multiorgan derangements, with respiratory predominance due to oxygen requirements
14	Classification driven predominantly by inflammatory derangements (temperature)
15	Classification driven by subtle multiorgan derangements, with SIRS predominance
16	Classification driven by subtle multiorgan derangements, with SIRS and respiratory predominance
17	Classification driven by subtle multiorgan derangements, with respiratory failure, SIRS, and liver predominance
18	Classification driven by subtle multiorgan derangements, with respiratory predominance and concomitant acidosis
19	Classification driven by subtle multiorgan derangements, with coagulopathy, metabolic (glucose), and inflammatory predominance
20	Classification driven by subtle multiorgan derangements, with inflammatory disequilibrium coagulopathy predominance
21	Classification driven by subtle multiorgan derangements, with respiratory, kidney, metabolic (acidosis), and hemodynamic (vascular tone) predominance
22	Classification driven by subtle multiorgan derangements, with liver, acidosis, and respiratory predominance
23	Classification driven by subtle multiorgan derangements, with inflammatory (temperature), respiratory failure, and liver failure
24	Classification driven by subtle multiorgan derangements, with inflammatory, respiratory, acidosis, and coagulopathy predominance

these approaches rely on deep learning. Their training pipelines, data sampling strategies, and hyperparameter optimization methods are often underspecified and unavailable for reproducibility. In contrast, Yang et al⁹ provide a well-documented training pipeline and data sampling approach, enabling precise benchmarking. Using Trust-MAPS to preprocess data within this pipeline improves the final utility score to 0.566, underscoring that Trust-MAPS can serve as a modular preprocessing tool for robust ML.

The concept of using projections on high-dimensional mathematical domain constraints has important implications in practice. First, the projection onto physical constraints can help notify practitioners if specific feature measurements have recurring errors. A model with the capabilities to identify erroneous clinical data could have different purposes depending on the stage of model development or implementation, to serve as a generalized preprocessing pipeline, or to identify potential data drifts or manual entry errors. Projections onto clinical constraints can enrich most algorithms with the required clinical context to recognize *potential* errors automatically. These outliers in EMR data can act as sources of systematic bias in ML models trained on this data, often disproportionately affecting certain patient populations or clinical scenarios and leading to unfair or inaccurate predictions. By enforcing clinically valid constraints through physical projections, Trust-MAPS actively mitigates these biases. Although the medico-legal implications of AI as diagnostic tools remains undefined,³⁵ our model's ability to compute projections on outlier data points could be treated as yet another quality control component in the healthcare infrastructure, particularly to guard against potential biases that may introduce generalization gaps during model development or implementation.

Second, trust-scores (or distance-to-normal measures) introduced in this work form a cohesive way to interpret key insights about patient physiology during the acute clinical trajectory. This approach can give clinicians or care providers the ability to quantify a patient's deviation from healthy physiology in a given interval of time, accounting for measured clinical values as well as temporal trends. By relying on fundamental physiological constraints rather than purely statistical normalization, our approach reduces the risk of learning spurious correlations that might reflect institutional practices or measurement biases rather than true clinical relationships. Trust-scores are an important meta-data that is predictive in the downstream ML pipeline. In addition, we found that trust-scores have clinical relevance across the different clusters, as detailed in Table 3. The top seven vitals and laboratory data with the highest variance across all clusters are TroponinI, PTT, Lactate, SaO₂, SOFA score, Bilirubin-direct, and FiO₂. Our results are consistent with the work of Zhao et al,³⁶ which reports PTT as a pivotal predictor of diagnosing sepsis early. The concept of trust-scores also aligns with contemporary clinical practice. The absolute value of a measurement is less important than how much it deviates from normal. This is demonstrated in Figure 6 in which a trust-score is the most important feature used to differentiate sepsis from non-sepsis in 17 of the 25 clusters. For instance, upon encountering a patient with multi-organ failure such as one experiencing significant liver failure (cluster 0, Figure 6), the degree to which an organ is failing is usually in direct proportion to how deviant the variable representing that organ is from "normal." A total bilirubin of 24 mg/dL is farther than 1.2 mg/dL (the upper limit of normal for total bilirubin) than 14 mg/dL. Thus, this projection-based framework can enable similar levels of targeted and precise clinical

management that may not be possible when considering models within the strong interpretation component afforded by such pipelines.

There are some limitations to this work. First, while we use the Euclidean distance metric to compute projections, alternative metrics may better suit certain constraints. Additionally, all constraints for clinical features were designed specifically for ICU patients in consultation with practicing clinicians; tailoring these constraints to specific patient subgroups could potentially enhance performance and enable precision medicine. Furthermore, all constraints are equally weighted in projection calculations, though assigning lower weights to frequently erroneous variables may improve error handling. Another limitation is that Trust-MAPS has been tested on a single dataset; extending this approach to additional datasets with potential domain shifts would help evaluate its generalizability. Finally, our pipeline can compute projections and run the ML model only after the full sub-patient data sequence (6 hours with 3-hour overlap) is available, introducing a 3-hour delay for real-time predictions. Shortening the sub-patient sequence duration could reduce this lag but might limit insights derived from temporal patterns in the data.

Leveraging projections to enhance data processing pipelines remains a relatively unexplored area of research. This work represents an initial investigation into this concept and demonstrates how incorporating clinically valid constraints can support health equity by ensuring models make predictions based on physiologically plausible data rather than artifacts or biases in the collection process. In future work, we aim to investigate other distance metrics (eg, KL divergence) for our projections pipelines. The constraints on clinical data can also be modified according to the illness being modeled, or the cohort being studied. Alternate formulations of the optimization objective could potentially lead to more faithful data correction results, for example, a stronger penalty on rate of change of measures variables. Furthermore, when creating sub-patients, we set the sepsis label for the *sub-patient* to “1” (yes) if the original patient ultimately developed sepsis in the time window considered for that sub-patient. In future work, it would be interesting to consider other design choices, for example, setting the *sepsis label* to be a continuous variable between 0 and 1, to show gradual progression to sepsis.

Conclusion

Predictive models that integrate data from various sources are highly sensitive to errors in data. Rapid identification of errors in clinical data is paramount to reducing alarm fatigue, improving robustness, and building unbiased models.³⁷

Trust-MAPS is a novel technique grounded in constrained optimization principles that we propose for enriching ML pipelines for clinical decision support. It can help with outlier detection and management in EMR data and can engineer a new set of features (trust-scores) with high predictive power for training ML models. We demonstrate the utility of this tool within the context of sepsis prediction for critically ill patients in the ICU. The proposed Trust-MAPS pipeline addresses the timely need to develop a new, context-aware framework that can dynamically identify nonsensical data, model the domain by translating clinical expertise into mathematical constraints, use projection-based methods to

automatically handle discrepancies in data to highlight trustworthy data as well as potential errors. We develop Trust-MAPS as a tool that can enrich any ML algorithm with clinical context, mitigating bias and increasing trustworthiness.

We finally remark that our projection approaches are general. The proposed concept of using projections to detect and correct erroneous data can be helpful in other clinical settings or non-clinical domains. We hope the positive results of this study inspire other researchers to use projections for data corrections and trust-scores to enhance their prediction pipelines.

Author contributions

Mehak Arora (Data curation, Formal Analysis, Software, Investigation, Validation, Visualization, Writing – original draft), Hassan Mortagy (Data curation, Formal Analysis, Software, Investigation, Validation, Visualization, Writing – original draft), Nathan Dwarshuis (Software, Formal Analysis), Jeffrey Wang (Formal analysis, Validation) and Philip Yang (Formal analysis, Validation), Andre Holder (Formal Analysis, Validation, Supervision, Writing – review & editing), Swati Gupta (Conceptualization, Methodology, Supervision, Funding Acquisition, Formal Analysis, Writing – review & editing), Rishikesan Kamaleswaran (Conceptualization, Methodology, Supervision, Funding Acquisition, Formal Analysis, Writing – review & editing)

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

The authors have no competing interests to declare.

Data availability

The data analyzed in this study is from the PhysioNet: Computing in Cardiology Challenge 2019 and is publicly available at <https://physionet.org/content/challenge-2019/1.0.0/>. The code used for our preprocessing and imputation pipeline, projection pipeline, and machine learning pipeline is documented and available at https://github.com/Kamaleswaran-Lab/ClinicalProjections_SepsisOnset.

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